

Shikun Sun

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My primary research focus is on diffusion models, with a specific interest in advancing new theoretical frameworks, enhancing controllable generation, exploring RL-based fine-tuning techniques, and improving computational efficiency.

Recent Work

- **Controllable Generation with Diffusion Models:** We apply multi-objective optimization to diffusion models to tackle real-world controllable generation tasks, leveraging the differentiability of the score function. Our work includes SDDM [ICML' 23] and the second part of Minimal Impact ControlNet [ICLR' 25], where we improve controllability under business-driven problem settings. We further revisit classifier guidance and propose SLCD [under review], a framework that preserves theoretical guarantees while enabling practical controllability.
- **RL-based Fine-tuning in Diffusion:** We reinterpret diffusion processes as Markov Decision Processes (MDPs) and enhance reward modeling to improve DPO, leading to DEPO [under review]. Additionally, we introduce proper entropy regularization into the diffusion MDP through EADPO [under review]. Finally, SLCD also proposes a KL-restrained, reward-based optimal solution that achieves strong results without the need for fine-tuning.
- **Theory-Implementation Consistency in Diffusion:** We revisit popular diffusion techniques from an SDE/ODE lens in Inner CFG [ICLR' 24], and enforce conservative properties in score functions via Minimal Impact ControlNet [ICLR' 25]. In Dual-Flow [under review], we propose a flow-based adversarial attack framework with adjoint-based improvement guarantees. The SLCD framework further offers classifier guidance while maintaining theoretical soundness, emphasizing consistency between theory and practice.

Education

- **Tsinghua University** Bachelor Degree
Computer Science and Technology 2017.09 - 2021.06
– Undergraduate Thesis: *Aesthetic Knowledge Guided Image Sentiment Transfer*
- **Tsinghua University** Second Bachelor Degree
Math and Applied Math 2018.09 - 2021.06
– Undergraduate Thesis: *Application of Invertible Neural Networks in Image Processing*
- **Tsinghua University** Pursuing Doctoral Degree
Multimedia Information Processing 2021.09 - now
– Expected to graduate in 2026.06
- **Cornell University** Visiting Scholar
DRL Theory and Application on Generative Models 2025.02 - 2025.07
– Visiting Scholar

Selected Award

- **Tsinghua University Academic Excellence Award** 2018
- **National Runner-up** *Tencent AI Arena National Open Competition, Honor of Kings* 2023

Grade-Point Average

• Bachelor of Computer Science and Technology	3.75
Bachelor of Math and Applied Math	3.26
Ph.D. in Multimedia Information Processing (by now)	3.89

Social Practice Activities

- Tsinghua University Bauhinia Volunteer Service Team *Member*

Publications

My research publications since 2023 have been centered on the theory and application of Diffusion Models.

- Ziyi Wang, Xingqi Wang, Zeyu Jin, Xiaohan Li, **Shikun Sun**, Jia Jia. *AI Carpet: Automatic Generation of Aesthetic Carpet Pattern*. In Proceedings of the 30th ACM International Conference on Multimedia (MM'22)
- Zijie Ye, Jia Jia, Haozhe Wu, Shuo Huang, **Shikun Sun**, Junliang Xing. *Salient Co-Speech Gesture Synthesizing with Discrete Motion Representation*. In Proceedings of 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP'23)
- **Shikun Sun**, Jia Jia, Haozhe Wu, Zijie Ye, Junliang Xing. *MSNet: A Deep Architecture using Multi-Sentiment Semantics for Sentiment-Aware Image Style Transfer*. In Proceedings of the 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP'23)
- **Shikun Sun**, Longhui Wei, Junliang Xing, Jia Jia, Qi Tian. *SDDM: Score-Decomposed Diffusion Models on Manifolds for Unpaired Image-to-Image Translation*. In Proceedings of the Fortieth International Conference on Machine Learning. (ICML'23)
- **Shikun Sun**, Longhui Wei, Zhicai Wang, Zixuan Wang, Junliang Xing, Jia Jia, Qi Tian. *Inner Classifier-Free Guidance and Its Taylor Expansion for Diffusion Models*. In Proceedings of the Twelfth International Conference on Learning Representations. (ICLR'24)
- Zixuan Wang, Jia Jia, **Shikun Sun**, Haozhe Wu, Rong Han, Zhenyu Li, Di Tang, Zhou Jiaqing, Jiebo Luo. *DanceCamera3D: 3D Camera Movement Synthesis with Music and Dance*. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024. (CVPR'24)
- Shuo Huang, **Shikun Sun**, Zixuan Wang, Xiaoyu Qin, Yanmin Xiong, Yuan Zhang, Pengfei Wan, Di Zhang, Jia Jia. *PlacidDreamer: Advancing Harmony in Text-to-3D Generation*. In Proceedings of the 32th ACM International Conference on Multimedia (MM'24)
- Zixuan Wang, Jiayi Li, Xiaoyu Qin, **Shikun Sun**, Songtao Zhou, Jia Jia, Jiebo Luo. *DanceCamAnimator: Keyframe-Based Controllable 3D Dance Camera Synthesis*. In Proceedings of the 32th ACM International Conference on Multimedia (MM'24)
- **Shikun Sun**, Min Zhou, Zixuan Wang, Xubin Li, Tiezheng Ge, Zijie Ye, Xiaoyu Qin, Junliang Xing, Bo Zheng, Jia Jia. *Minimal Impact ControlNet: Advancing Multi-ControlNet Integration*. In Proceedings of the thirteenth International Conference on Learning Representations. (ICLR'25)
- Renye Yan, Jikang Cheng, Yaozhong Gan, **Shikun Sun**, You Wu, Yunfan Yang, Ling Liang, JinLong Lin, Yeshuang Zhu, Jie Zhou, Jinchao Zhang, Junliang Xing, Yimao Cai, Ru Huang. *Entropy-Adaptive Diffusion Policy Optimization with Dynamic Step Alignment* (under review)
- **Shikun Sun**, Chengrui Wang, Min Zhou, Zixuan Wang, Xiaoyu Qin, Tiezheng Ge, Bo Zheng, Jia Jia. *DEPO: Enhancing E-commerce Image Background Generation with Short Trajectory Direct Expected Preference Optimization* (under review)

- Yixiao Chen*, **Shikun Sun***, Jianshu Li, Ruoyu Li, Zhe Li, Junliang Xing *Dual-Flow: Transferable Multi-Target, Instance-Agnostic Attacks via In-the-wild Cascading Flow Optimization* (under review)
- Owen Oertell*, **Shikun Sun***, Yiding Chen*, Jin Peng Zhou, Zhiyong Wang, and Wen Sun *Efficient Controllable Diffusion via Optimal Classifier Guidance*. (under review)